

CP020003

ARTIFICIAL INTELLIGENCE



Khon Kaen University · 2026

Week 12

Generative AI & the modern AI

From describing the world to inventing it



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https://github.com/kaopanboonyuen/CP020003_ArtificialIntelligence_2026s1



Today's map

Seven ideas, one thread: making instead of describing.

1

Two minds, two ways

LLM vs. diffusion

2

Text → image

Stable Diffusion v1.5

3

Image → text

Captioning with BLIP

4

The round trip

How much meaning survives

5

Inpainting

Editing only part of an image

6

One-step “turbo” generation

Distillation makes it instant

7

The 2026 frontier

What's newer than diffusion

1

Two minds, two ways to generate

Same goal — make something new — completely different process.

Large language model



committed → committed → committed → writing now ...

One token at a time, left to right, never undone.

A large language model is autoregressive. It picks one token, appends it, and conditions the next pick on everything before it — it can never reach back and change a word it already committed to.

Diffusion model



The whole canvas, revised together, many passes.

A diffusion model works the opposite way. It starts from pure noise and, at every step, revises the entire image at once — nothing is final until the last pass.

From the notebook — Section 1

We load Qwen2.5-0.5B-Instruct and print each token the instant it's generated — students watch the commitment happen live.

LLMs vs. diffusion, side by side

	Large language models	Diffusion models
Data type	Text — discrete tokens	Images / audio — continuous signal
How it generates	One token at a time, left to right	Whole canvas, refined over many passes
Training signal	Predict the next token	Predict the noise that was added
Analogy	Writing a sentence word by word, committing as you go	Sculpting a statue from noise, pass after pass
2026 frontier	Longer context, tool use, reasoning chains	Fewer steps (distillation), new math (flow matching)

2

Text → image: turning words into pixels

The flagship demo of generative AI — the one every student remembers.

Given a text prompt, generate a brand-new image that matches it. We run this with Stable Diffusion v1.5, guided by a text encoder that turns your words into the signal the denoiser follows at every step.

Two knobs control almost every generation

S

num_inference_steps

How many denoising passes it gets. More steps = more refinement, with diminishing returns and linear extra compute.

G

guidance_scale

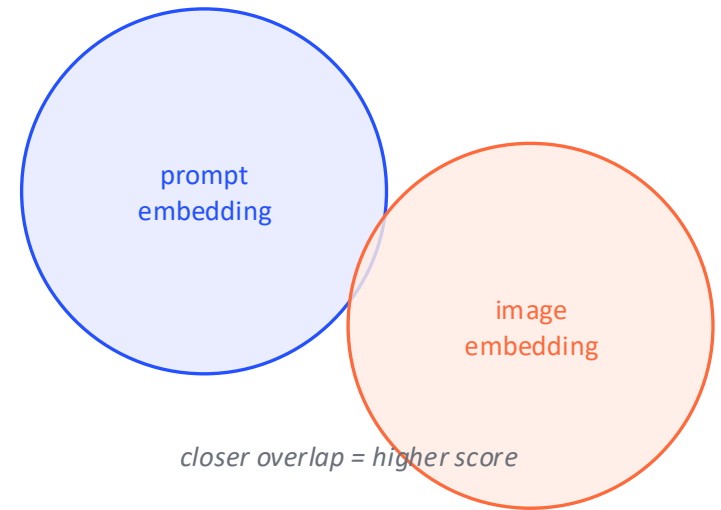
How strictly it follows your prompt vs. its own sense of a plausible image. Too low ignores you; too high over-sharpens.

From the notebook — Section 2.1 & 2.2

Three prompts, one seed — “a majestic red panda astronaut floating in space, digital art” and two more. Then a full sweep: steps [5, 15, 30] × guidance [1.5, 7.5, 15.0], nine images, one grid.

Scoring a “vibe” with an actual number

Looking at a grid of images and saying “that one looks better” isn't a metric — it's a feeling. CLIP Score encodes the prompt and the image into the same embedding space and measures how close they land: a real, comparable number.



openai/clip-vit-base-patch32

The exact checkpoint we use to score every image generated this week.

3

Image → text: describing what we made

The reverse direction — and the backbone of alt-text, search, and accessibility tools.

Given an image, generate a caption in natural language. We use BLIP, a small, fast captioning model, and point it straight at the art the class just generated — closing the loop between what diffusion made and how a language model reads it back.

From the notebook — Section 3

Salesforce/blip-image-captioning-base runs on every image from Section 2, turning pixels back into a sentence a language model can use.

1

Accessibility

Alt text for screen readers, generated automatically.

2

Search

Captions become searchable metadata for image libraries.

3

Moderation

A text summary a downstream system can reason about.

4

The round trip: image → text → image

A game that quietly teaches a deep lesson about meaning and compression.



Then we compare

Start with a prompt → Image A. Caption it → Prompt B. Regenerate from Prompt B → Image C. Compare A and C.

What usually survives

The main subject, the setting, and the broad style — a red panda astronaut stays a red panda astronaut.

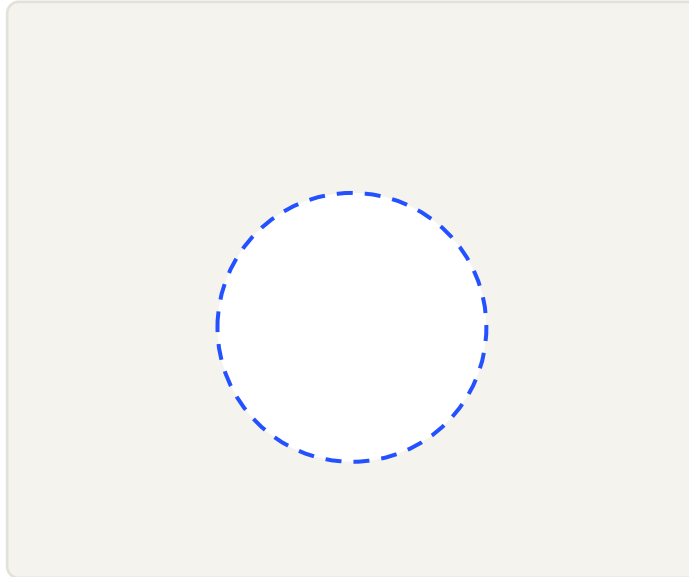
What usually drifts

Fine detail, exact composition, and mood — every translation between modalities is lossy, and here you can see it happen.

5

Inpainting: editing only part of an image

The technology behind “Magic Eraser” and “Generative Fill.”



white = edit this
black = leave alone

Given an image, a mask, and a new prompt, the model regenerates only the masked region while leaving everything outside it untouched — via `StableDiffusionInpaintPipeline`.

A good inpaint has to pass two checks at once:



CLIP score inside the mask

High similarity to the new prompt, measured only on the cropped region.



MSE outside the mask

Near zero difference from the original — stay quiet everywhere else.

From the notebook — Section 5.1

Match inside the circle, stay silent outside it — that's the whole job of an inpainting model.

6 Beyond classic diffusion: one-step “turbo” generation

The single biggest practical trend in generative AI since 2024.

Section 2 needed 25–30 denoising passes — the whole reason diffusion feels slow next to a single forward pass through a classifier. Distillation trains a fast student to jump straight to what a slow teacher reaches after dozens of steps.

A

Adversarial Diffusion Distillation

Powers stabilityai/sd-turbo. A discriminator pushes the few-step student toward the many-step teacher’s output.

L

Latent Consistency Models

Trained so any point on the noise-to-image path maps to the same final image — letting you leap most of the way there.

20–50 steps



1–4 steps

the difference between “wait ten seconds” and “instant.”

7

The 2026 frontier: what's newer than diffusion

This is where the LLM column and the diffusion column from Slide 4 start to merge.

1

Flow matching straightens the path

Stable Diffusion 3 and Flux train the model to follow a straight line from noise to image — fewer steps needed, baked in from training instead of bolted on after.

2

The U-Net retires, the transformer takes over

Diffusion Transformers (DiT) — the backbone behind Sora-style video models — replace the old convolutional denoiser with the same block that powers every LLM.

3

The autoregressive comeback

Some of the newest multimodal systems generate images token by token, left to right, inside the same transformer that writes the reply — no separate denoiser at all.

The big picture

2020: generative AI meant GANs. 2023: it meant diffusion. 2026: the frontier is unified multimodal transformers, where “LLM” and “image generator” are two settings of the same machine.

Wrap-up: the map of modern generative AI

Every task we ran this week, in one table.

Task	Model	Hugging Face ID	Output	Key metric
Text generation	Qwen2.5-0.5B-Instruct	<code>Qwen/Qwen2.5-0.5B-Instruct</code>	Token sequence	Human judgment
Text → image	Stable Diffusion v1.5	<code>stable-diffusion-v1-5/stable-diffusion-v1-5</code>	512×512 image	CLIP score
Image → text	BLIP (base)	<code>Salesforce/blip-image-captioning-base</code>	Caption	Human judgment
Masked editing	SD v1.5 Inpainting	<code>stable-diffusion-v1-5/stable-diffusion-inpainting</code>	Edited region	CLIP (in) / MSE (out)
Fast text → image	SD-Turbo	<code>stabilityai/sd-turbo</code>	512×512, 1–4 steps	CLIP vs. seconds

Try it yourself

Four small experiments that turn today's demos into your own results.

1

Break the guidance scale

Push `guidance_scale` above 20 on your own prompt. Describe the artifacts and connect them back to what the knob does.

2

Measure your own CLIP score

Write five prompts you think are hard for the model. Generate and score each — which failed lowest, and why?

3

Run the round trip on your own image

Pick a photo, caption it, regenerate from that caption, and list every detail that changed.

4

Swap in a newer backbone

Replace SD-Turbo with `sdxl-turbo` or a flow-matching checkpoint and repeat the speed-vs-quality comparison.

Four ideas to leave the room with

1

Discriminative looks; generative makes.

Everything before this week described the world. Everything this week invents it.

2

Autoregressive commits; diffusion revises.

One never looks back, the other never truly finishes until the last step.

3

Speed is a training problem, not just an engineering one.

Distillation and flow matching teach the model to need fewer steps in the first place.

4

The categories are merging.

The most modern systems don't pick a side of the LLM/diffusion table — they collapse it into one transformer.